

MR Thesis Questionnaire System

Survey Instrument Overview — Spanish Vocabulary Acquisition via Mixed Reality

This document describes every instrument used in the study session: the vocabulary Post-Test, the NASA-TLX workload questionnaire, the System Usability Scale (SUS), and the open-ended Qualitative questions. Each participant completes every instrument twice — once per visual condition (A and B) — except the Qualitative section, which is answered once at the end of the session.

1. Post-Test — Vocabulary Recall

Participants are shown 12 Spanish words per condition (24 words total) together with a picture and an audio playback button. Two question formats are interleaved to measure both **free recall** and **recognition**:

-  **PROGRESSIVE HINTS** — the participant types the Spanish word from memory. Up to two optional hints are available (first letter, then an English definition) and a "skip" option. No correctness feedback is shown.
-  **DIRECT MULTIPLE CHOICE** — the participant selects the Spanish word from four options (the correct word plus three distractors). No correctness feedback is shown.

Correctness is never revealed during the test — it is only aggregated afterward — because showing feedback mid-task would contaminate the embodied-memory measure being studied (Johnson-Glenberg, 2018; Ibrahim et al., 2018). A per-condition summary (accuracy at the free-recall level, at the recognition level, and overall) is shown to the participant right after each condition's 12th word.

1.1 Condition A — Spatial Labeling (Text Labels)

#	Spanish	English	Question type	Distractor options
1	teclado	keyboard	Progressive Hints	computadora, bolígrafo, cuaderno
2	taza	cup	Direct MC	plato, cuchara, leche
3	lentes	glasses	Progressive Hints	sombrero, teléfono, llavero
4	tenedor	fork	Direct MC	cuchara, plato, tijeras
5	cuchara	spoon	Progressive Hints	tenedor, taza, plato
6	cuaderno	notebook	Direct MC	bolígrafo, teclado, computadora
7	computadora	computer	Progressive Hints	teclado, teléfono, cuaderno
8	bolígrafo	pen	Direct MC	cuaderno, teclado, tijeras
9	sombrero	hat	Progressive Hints	gafas, llavero, juguete
10	teléfono	cellphone	Direct MC	computadora, gafas, llavero
11	llavero	keychain	Progressive Hints	gafas, teléfono, tijeras
12	tijeras	scissors	Direct MC	bolígrafo, llavero, tenedor

1.2 Condition B — Object Augmented (3D Models)

#	Spanish	English	Question type	Distractor options
1	pelota	ball	Progressive Hints	juguete, oso de peluche, manzana
2	leche	milk	Direct MC	queso, taza, plato
3	silla	chair	Progressive Hints	cama, lámpara, plato

#	Spanish	English	Question type	Distractor options
4	cama	bed	Direct MC	silla, lámpara, oso de peluche
5	zanahoria	carrot	Progressive Hints	manzana, queso, plato
6	plato	plate	Direct MC	taza, cuchara, queso
7	oso de peluche	teddy bear	Progressive Hints	juguete, pelota, cama
8	queso	cheese	Direct MC	leche, manzana, zanahoria
9	lámpara	lamp	Progressive Hints	silla, cama, microondas
10	manzana	apple	Direct MC	zanahoria, queso, leche
11	microondas	microwave	Progressive Hints	lámpara, computadora, plato
12	juguete	toy	Direct MC	pelota, oso de peluche, sombrero

2. NASA-TLX – Workload Questionnaire

Completed once per condition, right after its 12-word Post-Test. Six dimensions are each rated on a **1–21 slider scale**.

Dimension	Question shown to participant	Low anchor	High anchor
Mental Demand	How mentally demanding was the task?	Low (1)	High (21)
Physical Demand	How physically demanding was the task?	Low (1)	High (21)
Temporal Demand	How rushed or pressured did you feel?	Low (1)	High (21)
Performance	How successful were you in accomplishing the task?	Poor (1)	Perfect (21)
Effort	How hard did you have to work to accomplish your performance?	Low (1)	High (21)
Frustration	How insecure, discouraged, or frustrated did you feel?	Low (1)	High (21)

Scoring: the six ratings are averaged and expressed as a percentage of the maximum (21), giving a single 0–100 workload score per condition.

Hart, S. G., & Staveland, L. E. (1988). Development of NASA-TLX (Task Load Index). *Advances in Psychology*, 52, 139–183.

3. SUS – System Usability Scale

Completed once per condition. Ten standardized statements, each rated on a **1 (Strongly Disagree) to 5 (Strongly Agree)** scale. Wording and order are fixed and must not be modified, per the original instrument.

1. I think I would like to use this system frequently.
2. I found this system unnecessarily complex.
3. I thought the system was easy to use.
4. I think that I would need the support of a technical person to use this system.
5. I found the various functions in this system were well integrated.
6. I thought there was too much inconsistency in this system.
7. I would imagine that most people would learn to use this system very quickly.
8. I found the system very cumbersome to use.
9. I felt very confident using the system.
10. I needed to learn a lot of things before I could get going with this system.

Scoring: odd items contribute (rating – 1); even items contribute (5 – rating). The sum of all ten contributions is multiplied by 2.5, producing a 0–100 usability score.

Brooke, J. (1996). SUS: A retrospective. *Journal of Usability Studies*, 8(2), 29–40.

4. Qualitative Questions

Answered once at the end of the session (not per condition), as free-text responses up to 500 characters each.

1. Which visual interface did you prefer (Condition A: Text Labels or Condition B: 3D Models), and why?
2. Did you find the visual information in either condition overwhelming or distracting? Please explain.
3. How natural did the physical "poke" gesture feel? Did you experience any confusion on what to touch?
4. Which visual setup made it easier to focus on remembering the Spanish words, and why?
5. Did you experience any physical discomfort, visual fatigue, or technical issues?
6. Any other comments or suggestions?